ARTHROPLASTY · NEET-SS / FET / INI-SS

Bearing surfaces and wear

1. Tribology basics

- **Wear:** removal of material from articulating surfaces. **Mechanisms:**
  - **Adhesive** (commonest in metal-on-polyethylene: polymer transfers to the counterface and is pulled off),
  - **Abrasive** (hard asperities or particles plough the softer surface),
  - **Third-body** (trapped particles: cement, bone, metal debris),
  - **Fatigue/delamination** (subsurface cracking; important in **TKA**, low-conformity, oxidised polyethylene),
  - **Corrosive** (fretting at modular junctions).
- **Modes of wear (McKellop): Mode 1** intended bearing surfaces; **Mode 2** a primary surface against a non-intended secondary surface (e.g. head against shell after liner wear-through); **Mode 3** third-body particles between primary surfaces; **Mode 4** two non-intended surfaces rubbing (back-side wear of a liner against the shell, neck against the cup rim, **trunnion corrosion**).
- **Lubrication: fluid-film** (ideal; surfaces fully separated: ceramic-on-ceramic and well-designed metal-on-metal with large heads and low clearance), **boundary** (surfaces in contact, lubricant adsorbed: metal-on-polyethylene), **mixed**. Lambda ratio (fluid-film thickness / combined surface roughness) **over 3 = full fluid-film**.
- **Linear wear** (penetration of the head into the liner, mm/year) vs **volumetric wear** (mm^3/year: **determines particle load and osteolysis**). **Volumetric wear increases with head size** ($V = \pi r^2 w$, Charnley); **linear wear decreases with larger heads**.
- **Osteolysis threshold** for conventional polyethylene: **about 0.1 mm/year linear wear** (Dumbleton); above this, osteolysis is common.

2. Biology of particle disease (osteolysis)

- **Wear particles (0.1-1 micron polyethylene most bioactive)** are phagocytosed by **macrophages**, which release **TNF-alpha, IL-1, IL-6, PGE2**,

upregulate **RANKL** on osteoblasts/stromal cells: **osteoclast activation** and bone resorption. The **effective joint space** (Schmalzried) lets particles migrate along the implant-bone interface.

- **Clinical:** often **silent** until large lesions, loosening or fracture; **serial radiographs**, **CT** for extent (pelvic lesions underestimated on radiographs).
- **Treatment: liner exchange and bone grafting** of lytic lesions with retained well-fixed components (risk of dislocation after isolated liner exchange: use larger heads), or full revision for loose components.

3. Polyethylene

- **UHMWPE (ultra-high molecular weight polyethylene):** semi-crystalline polymer. **Gamma irradiation** (sterilisation) breaks chains forming **free radicals**: in air, these **oxidise** (embrittlement, **subsurface white band**, delamination, worse with **shelf ageing**).
- **Sterilisation:** gamma in **inert gas/vacuum** packaging, **ethylene oxide** or **gas plasma** (no cross-linking, no free radicals).
- **Highly cross-linked polyethylene (HXLPE):** irradiated at **5-10 Mrad** to cross-link chains (**much lower adhesive/abrasive wear**: **wear reduced about 80-90%**, dramatic fall in osteolysis and revision for wear in THA); **post-irradiation thermal treatment** removes free radicals: **remelting (above the melting point, about 137 C: eliminates free radicals but reduces crystallinity, fatigue strength and fracture toughness)** or **annealing (below melting: preserves mechanical properties but leaves some free radicals: in-vivo oxidation)**. **Vitamin E (alpha-tocopherol)-infused/blended HXLPE** quenches free radicals without remelting.
- **Cross-linking reduces fatigue strength and toughness**: concern in **TKA (high contact stress, fatigue wear)** and thin liners, **rim/locking mechanism fractures** with thin liners and large heads or malposition.
- **Polyethylene thickness: at least 6-8 mm** in TKA (thinner = high contact stress, delamination).
- **Compression-moulded (direct)** polyethylene has better wear performance than **ram-extruded + machined** stock in TKA; **Hylamer** (failed); **calcium stearate**-free resins.

- **TKA wear:** fatigue-dominated (**delamination, pitting**) because of **low conformity**, rolling/sliding, high contact stresses; **backside wear** of modular inserts (micromotion against the tray) contributes to tibial osteolysis.

4. Metal-on-polyethylene (MoP)

- **Cobalt-chromium or ceramic heads on HXLPE**: the commonest and most proven bearing in THA (excellent long-term registry survival).
- **Head size: 32-36 mm** heads with HXLPE balance dislocation risk and wear; very thin liners for large heads risk fracture.

5. Ceramics

- **Alumina (Al_2O_3)** and **zirconia-toughened alumina (BIOLOX delta, ZTA)**: **hardest, most scratch-resistant, very low wear, inert particles, excellent wettability (hydrophilic: fluid-film lubrication)**, **lowest wear of all bearings (ceramic-on-ceramic)**.
- **Problems: brittleness** (fracture: much rarer with delta ceramics, about 0.002-0.02%), **squeaking** (about 1-3%: associated with **cup malposition (edge loading)**, **stripe wear**, impingement), **chipping on insertion** (malseating), **limited neck lengths**, cost.
- **Ceramic head fracture:** revise to a **ceramic head** (not metal: **third-body ceramic particles** cause catastrophic metal head wear) with full synovectomy; use a new taper sleeve if the trunnion is damaged.
- **Ceramic-on-polyethylene:** reduces polyethylene wear vs metal heads (smoother, harder, more wettable), avoids taper corrosion (trunnionosis) of CoCr heads.
- **Oxidised zirconium (Oxinium):** zirconium alloy with a **ceramic oxide surface layer**: low wear against polyethylene with metal toughness; **avoid against a damaged surface**: if the thin layer is breached (e.g. after dislocation and closed reduction over the cup rim), the soft substrate wears rapidly.
- **Old zirconia (yttria-stabilised) heads** suffered **phase transformation** (Prozyr recall 2001).

6. Metal-on-metal (MoM)

- **Cobalt-chromium on cobalt-chromium**; large heads possible (hip resurfacing: **Birmingham Hip Resurfacing**; large-head THA). Low volumetric wear in ideal conditions but **nanometre-sized metal particles** (huge numbers) and **metal ions (cobalt, chromium)**.
- **Adverse reactions to metal debris (ARMD) / adverse local tissue reaction (ALTR): pseudotumours** (solid or cystic masses), **ALVAL** (aseptic lymphocyte-dominated vasculitis-associated lesion: type IV delayed hypersensitivity), **metallosis**, **soft-tissue necrosis** (abductor destruction). **Systemic cobalt toxicity:** cardiomyopathy, neuropathy, hypothyroidism, visual/hearing loss.
- **Risk factors: female sex**, **small component size**, **steep cup inclination (over about 50-55 degrees: edge loading)**, **specific designs (ASR recall 2010)**, large-head MoM THA (trunnion corrosion), bilateral implants, renal impairment.
- **Surveillance (MHRA/FDA): blood cobalt and chromium levels** (concern **over about 7 µg/L (ppb)**; some use 2-4 µg/L for unilateral resurfacing as threshold for closer monitoring), **MARS MRI** (metal artefact reduction sequence) or ultrasound for pseudotumours.
- **Revision** for symptomatic patients, rising ions, enlarging pseudotumours or osteolysis; revise to a **ceramic or polyethylene** bearing. **Outcomes of revision for ALTR are worse** than for other aseptic causes.
- **Hip resurfacing: young, active men with large femoral heads** (best results; **avoid in women, small heads, osteoporosis, large cysts, AVN, DDH**). **Femoral neck fracture** (commonest early failure; **notching** of the neck, varus positioning).
- **Metal hypersensitivity** in TKA (nickel) is debated (rarely a true cause of failure; diagnosis of exclusion).

7. Taper corrosion (trunnionosis)

- **Mechanically assisted crevice corrosion** at the **head-neck (and modular neck-stem) junction**: micromotion disrupts the passive oxide layer: fretting + crevice corrosion: **cobalt release**, **ALTR even with metal-on-polyethylene**.

- **Risk factors: large CoCr heads** (larger moment arm/torque), **long necks/high offset**, **dissimilar metals (CoCr head on titanium stem)**, small/short trunnions, **dual-taper modular necks** (recalled designs), poor taper assembly (blood/fluid on the taper, inadequate impaction).
- **Diagnosis: raised cobalt >> chromium** (Co:Cr ratio over about 3-5 suggests taper corrosion rather than bearing wear), **MARS MRI**.
- **Treatment: revise the head to a ceramic head with a titanium sleeve**, change liner, debride; revise the stem if the trunnion is badly damaged.

8. Choosing a bearing (summary)

Bearing	Wear	Particles	Main concerns	Typical use
MoP (CoCr on HXLPE)	Low (with HXLPE)	Polyethylene (osteolysis)	Taper corrosion with large CoCr heads	**Most patients**
Ceramic-on-HXLPE	Lower	Polyethylene	Cost; ceramic fracture (rare)	Younger/active patients; avoid taper issues
Ceramic-on-ceramic	**Lowest**	Ceramic (inert)	**Squeaking, fracture, chipping**, cup position sensitive	Young active patients
MoM	Low (volumetric) but ions	**Metal nanoparticles/ions**	**ALTR/pseudotumour, systemic cobalt**	Largely abandoned except selected resurfacing

- **Hardness (decreasing):** ceramic > CoCr > stainless steel > titanium > polyethylene. **Titanium is never used as a bearing surface** (poor wear resistance, notch sensitivity).
- **Wettability (hydrophilicity):** ceramics best.

Exam pearl: **Volumetric wear determines osteolysis; larger heads = more volumetric, less linear wear (with conventional PE). Osteolysis threshold about 0.1 mm/year. Particles of 0.1-1 micron activate macrophages, TNF-alpha, RANKL.**

Exam pearl: **HXLPE reduces THA wear about 80-90%; remelting removes free radicals but lowers fatigue strength; annealing keeps strength but leaves free radicals; vitamin E is the alternative.** TKA wear is fatigue/delamination (conformity and thickness at least 6-8 mm).

Exam pearl: **Ceramic-on-ceramic: lowest wear; squeaking and fracture; revise a broken ceramic head with a new ceramic head. MoM: pseudotumour/ALVAL, women, small sizes, steep cups; check cobalt/chromium and MARS MRI. Trunnionosis: cobalt much higher than chromium.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.